

Statics is the branch of Mechanics that deals with Solids at rest under the influence of forces i.e. it is the study of motionless system or systems that move with constant velocity. It is a special case of dynamics in which the acceleration is equal to zero. It is concerned with forces in equilibrium.

Mechanics is the physical science which deals with the effect of forces on objects. It is the science that considers the motion of bodies under the action of forces.

SCALAR AND VECTOR QUANTITIES

A scalar quantity is a quantity that has only magnitude with no direction.

Examples of scalar quantities are mass, length, energy, density, speed, volume, temperature, current, time etc.

A vector quantity is a quantity that has both magnitude and direction.

Examples of vector quantities include: displacement, velocity, acceleration,

impulse, Momentum, Moment, electric field, force

PARTICLES AND RIGID BODIES

A Particle is an object that has

infinitely small volume (occupies negligible space) but has a mass which can be considered to be concentrated at a point. It is a body whose size does not influence its response to forces acting on it. It can also be defined as a body whose size in a given physical situation is not significant in the analysis of its response to forces that acts on it.

In a physical situation in which the size or rotation of a body can not be ignored, the body can not be treated as a particle.

Example is the throwing of a shot; the shot can be modeled as a particle subject to three forces

1. from the hand H ,

2. from the chin C and.

3. weight due to gravity W .

A rigid body is a body that does not deform under the action of forces. In a rigid body, the distance between any two particles remains constant, i.e. the size and shape of the body do not change. A rigid body exhibits no relative deformation between two points when it is acted upon by a force system. If the deformation of the body is very small and has little effect on its motion or equilibrium, that deformation can be ignored when considering the body's motion or equilibrium; the body is assumed to be rigid and is referred to as a rigid body. Each point in a rigid body is always at a constant distance from any other point in the body. An example of a body that is not rigid is a car that collides with a wall and a large deformation of the car occurs which leads to subsequent motion.

An elastic body undergoes deformation but regains its original shape after removal of the external forces.

3TH FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF MECHANICAL ENGINEERING
1ST AND 3RD SEMESTER EXAMINATIONS 2022/2023 ACADEMIC SESSION
COURSE CODE/TITLE: MEC 112/107/111, MECHANICAL ENGINEERING

PROGRAMME: ND1, (MECHANICAL ENGG, COMP. ENGG & AGRIC.ENGG.
SCIENCE I
TECHNOLOGY)

TIME ALLOWED:

3 HOURS

INSTRUCTION:

ANSWER ANY FIVE (5) QUESTIONS

ALL QUESTION CARRY EQUAL MARKS

QUESTION 1

- (a) What is statics? (3 Marks)
- (b) State Lami's theorem. (3 Marks)
- (c) A bag of mass **16kg** hangs from two cords of length **12m** and **15m**. The other ends of the cords are attached to a vertical board to the points at the same level and **20m** apart. Calculate the tension in the two strings. (14 Marks)

QUESTION 2

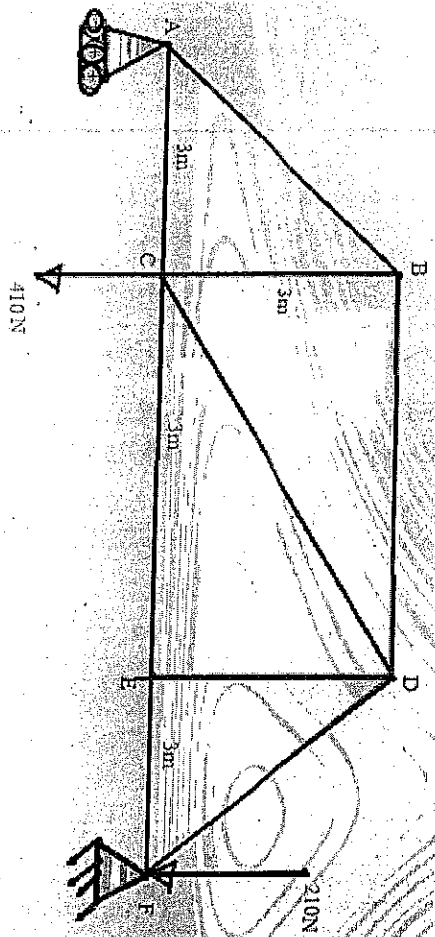
- (a) Define force and state its characteristics. (5Marks)
- (b) State the law of parallelogram of forces. (3 Marks)
- (c) Forces of **70N** act horizontally on a particle while the second force **90N** act at an angle of **50°** from horizontal. Find the resultant and the angle it makes with horizontal using parallelogram law. (12 Marks)

QUESTION 3

- (a) Define moment of a force and state the principle of moment. (4 Marks)
 - (b) Distinguish between contact and field force and give three (3) examples of each. (4 Marks)
 - (c) Forces equal to **12P**, **16P** and **18.17P** acting on a particle are in equilibrium, calculate the angle between the forces **12P** and **18.17P** (6 Marks)
- (10 Marks)

QUESTION 4

- (a) Define the following terms (i) Hydrostatics Pressure (ii) Relative density. (4 Marks)
- (b) State Archimedes principle. (4 Marks)
- (c) The figure below shows the forces acting at a point o. Find the magnitude and the direction of the resultant. (12 Marks)



MEC111

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF MECHANICAL ENGINEERING
1ST SEMESTER EXAMINATION 2012/2013 ACADEMIC SESSION

COURSE TITLE/CODE: MECH. ENG'G SCIENCE (STATICS) MEC 111
PROGRAMME: ND 1 (AGRIC ENG'G)
TIME ALLOWED: 3 HOURS
INSTRUCTIONS: ANSWER ANY FIVE QUESTIONS.
ALL QUESTIONS CARRY EQUAL MARKS.

QUESTION ONE

- a) Define the term angle of repose. (2marks)
- b) Give four (4) engineering applications of the theory of angle of repose. (4marks)
- c) A mass of 12kg is placed on a plane inclined at an angle of 30° to the horizontal. What force parallel to the plane is required to hold the mass at rest (take $\mu = 0.4$ & $g = 9.81\text{m/s}^2$) (14marks)

QUESTION TWO

- a) What is a lubricant? (3marks)
- b) State Lami's theorem. (3marks)
- c) A mass of 10kg hangs from two light springs 9m and 12m long tied to two points at the same level 15m apart. Find the tension in each string. (14marks)

QUESTION THREE

- a) Define moment of a force. (2marks)
- b) State the parallelogram law of vectors. (3marks)
- c) A uniform beam rest in a horizontal position supported at a point distance 0.7m from one end and carrying a mass of 12kg suspended from this end. The reaction at a support is 310N. Determine the mass and length of the beam. (15marks)

QUESTION FOUR

- a) Explain the following. i) Static friction. ii) Dynamic friction (4marks)
- b) State any four (4) laws of friction (4marks)
- c) A block of mass 20kg rests on a horizontal ground (coefficient of friction 0.3). If the block is pulled with a force of 100N at an angle of 60° to the horizontal, with what acceleration does it moves? (take $g = 10\text{m/s}^2$) (12marks)

QUESTION FIVE

- a) Define the term truss. (2marks)
- b) Explain the following types of truss i) perfect truss ii) imperfect truss (4marks)
- c) Use the method of joint to determine the forces along AB, BC & FD as shown in fig below. Indicate whether the forces are tensile or compressive. (14marks)

ACTUARIA

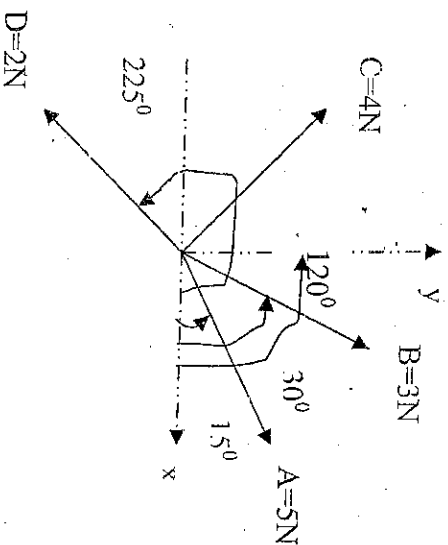
THE FEDERAL POLYTECHNIC MOBI
DEPARTMENT OF MECHANICAL ENGINEERING
1ST AND 3RD SEMESTER EXAMINATIONS 2010/2011 ACADEMIC SESSION
COURSE CODE/TITLE: MEC 111 MECHANICAL ENGG.SCIENCE.
PROGRAMME: NDI (MECHANICAL ENGINEERING)
TIME ALLOWED: 3 HOURS
INSTRUCTION: ANSWER ANY FIVE QUESTIONS.

Q.1.

- What is statics? (2mks)
- Differentiate between scalar and vector quantities and give four (4) examples of each. (6mks)
- A stone of mass 30kg rests on a horizontal ground, if the stone is pulled with a force of 120N at an angle of 60° to the horizontal with what acceleration does it moves? NOTE ($\mu = 0.3$ and $g = 10\text{m/s}^2$). (12mks)

Q.2.

- Define moment of a force. (2mks)
- State Lami's theorem. (2mks)
- The figure below shows the forces acting on a point O. Find the magnitude and the direction of the resultant. (16mks)



Q.3.

- Define friction and state four (4) areas where the application of friction is important. (4mks)
- (i) Give two (2) disadvantages of friction (2mks)
(ii) Give two (2) ways of reducing friction (2mks)
- A mass of 5kg hangs from two tight strings of length, 3m and 4m from two points at the same level and 5m apart. Find the tensions in the strings. (Assume $g = 10\text{m/s}^2$). (12mks)

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF MECHANICAL ENGINEERING

1ST AND 3RD SEMESTER EXAMINATION 2015/2016 ACADEMIC SESSION

COURSE CODE/ TITLE: MEC 111/ MECH. ENG-G SCIENCE (STATICS)

PROGRAMME: ND 1 (AGRIC ENG-G)

TIME ALLOWED: 3 HOURS

INSTRUCTIONS: ANSWER ANY FIVE QUESTIONS.

ALL QUESTIONS CARRY EQUAL MARKS.

QUESTION ONE

- (a) Define force. 2 marks
- (b) Distinguish between Contact and field force and give two examples of each. 6 marks
- (c) A truck of mass 2100kg is to be hauled up an inclined of 1 in 14. What constant force applied Parallel to the plane, is required to move it at a constant acceleration of 8 m/s^2 ? Take $g=9.81\text{ m/s}^2$. 14 marks

QUESTION TWO

- (a) Explain the term resultant. 2 marks
- (b) State the law of parallelogram of forces. 3 marks
- (c) A force of 75N act horizontally on a particle while the second force 95N act at an angle of 47° from horizontal. Find the resultant and the angle it makes with horizontal using the parallelogram law. 15 marks.

QUESTION THREE

- (a) Define statics. 2 marks
- (b) State Lami's theorem. 3 marks
- (c) A body of mass 5kg hangs from two cords of length 3.5m and 4.5m. The other ends of the cords are attached to a vertical board to points at the same level and 5.5m apart. Calculate the tension in the two strings. 15 marks

QUESTION FOUR

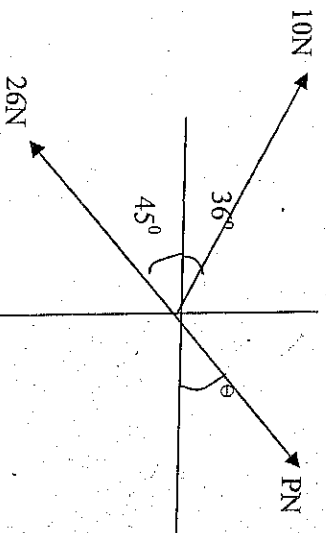
- (a) Differentiates between concurrent and coplanar forces. 2 mark
- (b) State the principles of moment. 3 marks
- (c) A uniform beam rest in horizontal position supported at a point distance 0.7m from one end and carry a mass of 15kg suspended from this end. The reaction on the support is 310N. Determine the mass and length of the beam. Take $g=9.81\text{ m/s}^2$. 15 marks

QUESTION FIVE

- Define the term Friction. 2 marks
- Give four (4) advantages and disadvantages of friction. 8 marks
- A mass of 15kg is placed on a plane inclined at an angle of 30° to the horizontal. What force parallel to the plane is required to hold the mass at rest? Take $\mu=0.4$ $g=9.81\text{m/s}^2$. 10 marks

QUESTION SIX

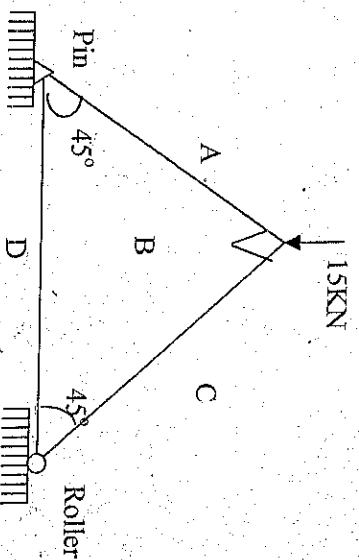
- Differentiate between Scalar and vector quantities and give four example of each 6 marks
- In the diagram shown in figure 1 below forces of magnitude 10N, 26N and PN are acting on a body in the direction indicated. If the body is in equilibrium, find the magnitude and direction of the force P. 14 marks



14 marks

QUESTION SEVEN

- Define the term framed structure. 3marks
- Explain the following terms: 3marks
 - Deficient frames (truss)
 - Redundant truss
- Calculate the forces in all members of truss shown in figure 2 below using the one fluid of joints 3marks



14marks

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF MECHANICAL ENGINEERING
1ST AND 3RD SEMESTER EXAMINATION 2014/2015 ACADEMIC SESSION

COURSE TITLE/CODE: MECH. ENG/G SCIENCE (STATICS) MEC 111
PROGRAMME: ND 1 (AGRIC. ENG/G)
TIME ALLOWED: 3HOURS
INSTRUCTIONS: ANSWER ANY FIVE QUESTIONS.
ALL QUESTIONS CARRY EQUAL MARKS.

QUESTION ONE

- a) Define moment of a force 2marks
- b) State the principle of moment 3marks
- c) A uniform beam of length 4m, weight 200N and holds a 45kg weight. Find the magnitude of the force exerted on the beam by the two supports at its ends. 15marks

QUESTION TWO

- a) What is Statics? 2marks
- b) Differentiate between scalar and vector quantities and give four examples of each. 6marks
- c) Forces equal to 11p, 15p and 17.17p acting on a particle are in equilibrium; calculate the angle between forces 11p and 17.17p. 15marks

12marks

QUESTION THREE

- a) Define friction and state four (4) areas where the application of friction is important. 6marks
- b) Give any four (4) ways of reducing friction. 2marks
- c) A block of mass 78kg rest on a horizontal floor. What force is required just to make the block move when:
 - i) Pulling horizontally. 12marks
 - ii) Pulling at an angle of 60° to the horizontal. (Take $\mu = 0.4$, $g = 10\text{m/s}^2$).

QUESTION FOUR

- a) Exams the following terms:
 - i) Resultant 4marks
 - ii) Equilibrant. 3marks
- b) State Lami's theorem. 3marks
- c) A bag of mass 15kg hangs from two cords of length 12m and 15m. The other ends of the cords are attached to a vertical board to the points at the same level and 20m apart. Calculate the tension in the two strings. 13marks

QUESTION FIVE

- a) Define force. 2marks
- b) State the law of parallelogram of forces. 3marks
- c) Determine the magnitude and direction of the resultant of the forces acting at the same point and magnitude 64kN and 96kN analytically displaced relatively by 60° to each other. 15marks

QUESTION SIX

- a) Name any four (4) types of dynamics friction. 2marks
- b) Give four (4) engineering applications of the theory of angle of repose. 4marks

Tofa Tawana